

Utility of Deep Learning Super-Resolution in the Context of Osteoarthritis MRI Biomarkers

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Background: Super-resolution is an emerging method for enhancing MRI resolution; however, its impact on image quality is still unknown.

Purpose: To evaluate MRI super-resolution using quantitative and qualitative metrics of cartilage morphometry, osteophyte detection, and global image blurring.

Study Type: Retrospective.

Population: In all, 176 MRI studies of subjects at varying stages of osteoarthritis.

Field Strength/Sequence: Original-resolution 3D double-echo steady-state (DESS) and DESS with 3x thicker slices retrospectively enhanced using super-resolution and tricubic interpolation (TCI) at 3T.

Assessment: A quantitative comparison of femoral cartilage morphometry was performed for the original-resolution DESS, the super-resolution, and the TCI scans in 17 subjects. A reader study by three musculoskeletal radiologists assessed cartilage image quality, overall image sharpness, and osteophytes incidence in all three sets of scans. A referenceless blurring metric evaluated blurring in all three image dimensions for the three sets of scans.

Statistical Tests: Mann-Whitney *U*-tests compared Dice coefficients (DC) of segmentation accuracy for the DESS, super-resolution, and TCI images, along with the image quality readings and blurring metrics. Sensitivity, specificity, and diagnostic odds ratio (DOR) with 95% confidence intervals compared osteophyte detection for the super-resolution and TCI images, with the original-resolution as a reference.

Results: DC for the original-resolution ($90.2 \pm 1.7\%$) and super-resolution ($89.6 \pm 2.0\%$) were significantly higher ($P < 0.001$) than TCI ($86.3 \pm 5.6\%$). Segmentation overlap of super-resolution with the original-resolution (DC = $97.6 \pm 0.7\%$) was significantly higher ($P < 0.0001$) than TCI overlap (DC = $95.0 \pm 1.1\%$). Cartilage image quality for sharpness and contrast levels, and the through-plane quantitative blur factor for super-resolution images, was significantly ($P < 0.001$) better than TCI. Super-resolution osteophyte detection sensitivity of 80% (76–82%), specificity of 93% (92–94%), and DOR of 32 (22–46) was significantly higher ($P < 0.001$) than TCI sensitivity of 73% (69–76%), specificity of 90% (89–91%), and DOR of 17 (13–22).

Data Conclusion: Super-resolution appears to consistently outperform naïve interpolation and may improve image quality without biasing quantitative biomarkers.

Level of Evidence: 2

Technical Efficacy: Stage 2

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OSTEoarthritis (OA) affects over 30 million adults in the United States and is a leading source of pain.¹ Knee OA accounts for ~80% of the burden of the disease.² While radiography currently serves as the standard for OA diagnoses, it is insensitive to the multitude of soft tissues affected in knee OA activity and can only detect latestage OA changes.³ Magnetic resonance imaging (MRI) is a common tool for studying soft-tissue changes caused by OA and may provide potential biomarkers sensitive to early OA.^{4,5}

Within MRI, there is great interest in improving resolution for detecting subtle anatomical defects occurring in early OA activity and for generating high-resolution 3D images. However, high-resolution MRI is primarily limited by long scan times, which can increase the time and costs of large research studies. For example, the Osteoarthritis Initiative (OAI) included an 11-minute high-resolution double-echo steady-state (DESS) pulse sequence that was used to scan over 25,000 subjects at varying timepoints.⁶ Traditional MRI acceleration methods such as parallel imaging may come at the expense of noisier images due to g-factor limitations, while compressed sensing can come at the expense of long reconstruction durations and tradeoffs between artifacts and blurring due to the use of empirical regularization parameters.^{7–9} The use of 3D fast spin echo sequences is promising to expedite image acquisition; however, such images may be susceptible to image blurring due to k -space modulation over long echo train lengths.¹⁰

Image super-resolution (SR) includes an established family of computer vision techniques that operate in image space by directly transforming low-resolution images into higher-resolutions, using a variety of algorithms.¹¹ Convolutional neural networks (CNNs) and deep-learning-based SR may provide an alternative technique to reduce MRI acquisition time without the tradeoffs of conventional acceleration methods.^{12,13} Recently, SR has been used on DESS acquisitions from the OAI with promising results; however, the evaluation was performed only using image quality metrics such as structural similarity (SSIM), mean-square-errors (MSE), and peak signal-to-noise ratio (pSNR).^{14,15} While these metrics are common loss functions for CNN training, they do not always correlate with perceived image quality.^{16,17}

Previous applications of DESS SR have only been optimized for heuristic image quality metrics and the accuracy of regional T_2 relaxation time measurements, respectively.^{18,19} However, the impact of SR on the perceptual image quality and its impact on resolution-dependent techniques such as quantitative morphological analysis (through segmentation) and abnormality detection have yet to be evaluated. Thus, it remains unclear whether the same biomarkers can be obtained with SR DESS images as compared with the original high-resolution DESS images. Specifically, the DESS sequence in the OAI was primarily used to evaluate articular cartilage morphometry and osteophytes, which are osteo-cartilaginous protrusions developing on the margins of osteoarthritic joints.²⁰ Variations in both cartilage and osteophytes are known hallmarks of OA

activity.^{21,22} Consequently, the purpose of this study was to determine the impact of SR on potential imaging biomarkers of OA progression.

Materials and Methods

Subject Population

We used publicly available DESS data from the OAI, which is a longitudinal observational cohort study investigating the natural history of and risk factors for knee OA. The scan parameters for DESS were as follows: field of view (FOV) = 14 cm, matrix = 384×307 (zero-filled to 384×384), spatial resolution = $0.36 \text{ mm} \times 0.45 \text{ mm}$, echo time / repetition time (TE/TR) = 5/16 msec, slice thickness = 0.7 mm, and number of slices = 160.⁶ In all, 124 3D DESS image volumes were used for network training, 35 for validation, and 17 for testing. The distribution of subjects with varying Kellgren–Lawrence grades (KLG) of knee OA severity was maintained approximately equally in the 124 training datasets (3 KLG-1, 41 KLG-2, 71 KLG-3, 9 KLG-4), 35 validation datasets (1 KLG-1, 12 KLG-2, 20 KLG-3, 2 KLG-4), and 17 testing (6 KLG-2, 10 KLG-3, 1 KLG-4) datasets. KLG grades were obtained directly from the public OAI readings database, where they were defined through centralized readings of fixed flexion radiographs by two readers (a radiologist and rheumatologist).²³ In case of disagreement, a consensus read was performed with a third reader. The placement of subjects into the training, validation, and testing groups was performed in fully randomized manner. 3D DESS images from all 17 testing subjects for all three image sets (original high-resolution, DeepResolve SR, and tricubic interpolation) were used in the quantitative and qualitative experiments described below.

Super-Resolution

We utilized a 3D CNN entitled DeepResolve to transform low-resolution images into higher-resolution images by learning a difference image (the residual image) between the two.¹³ Given a set of low-resolution ($x^{(l)}$) and high-resolution ($y^{(h)}$) images, and residuals ($r^{(l)} = x^{(l)} - y^{(h)}$); DeepResolve was used to learn residual transformations $\hat{r} = f(x)$ during training. The training was subject to an L2-loss between the high-resolution and SR images. During inference, a SR image $\hat{y} = x + f(x)$ was estimated using a summation of the underlying low-resolution image using (x) and the estimated residual image $f(x)$.

The original DESS acquisition included slices with a thickness of 0.7 mm. DeepResolve utilized DESS slices with a thickness of 2.1 mm. First, low-resolution DESS slices with $3\times$ thicker slices (2.1 mm) were generated using a 48th-order antialiasing filter. Second, the commonly used approach of tricubic interpolation (TCI) was used to interpolate the thicker slices to the same slice locations of the original high-resolution sequence, thereby creating paired low- and high-resolution images.¹³ The $3\times$ resolution downsampling factor was chosen from previous findings, which demonstrated that using higher downsampling factors with the current SR network may lead to excessive image blurring that may not be recovered using DeepResolve.¹³ Overall, the TCI images were the input to DeepResolve, whose output were the super-resolved images. DeepResolve training was performed using $32 \times 32 \times 32$ patches using a CNN that utilized 20 convolution operators with 64 filters, each of kernel size $3 \times 3 \times 3$ and rectified linear unit activations (Fig. 1a).²⁴ The network structure was described in detail previously.^{13,25}

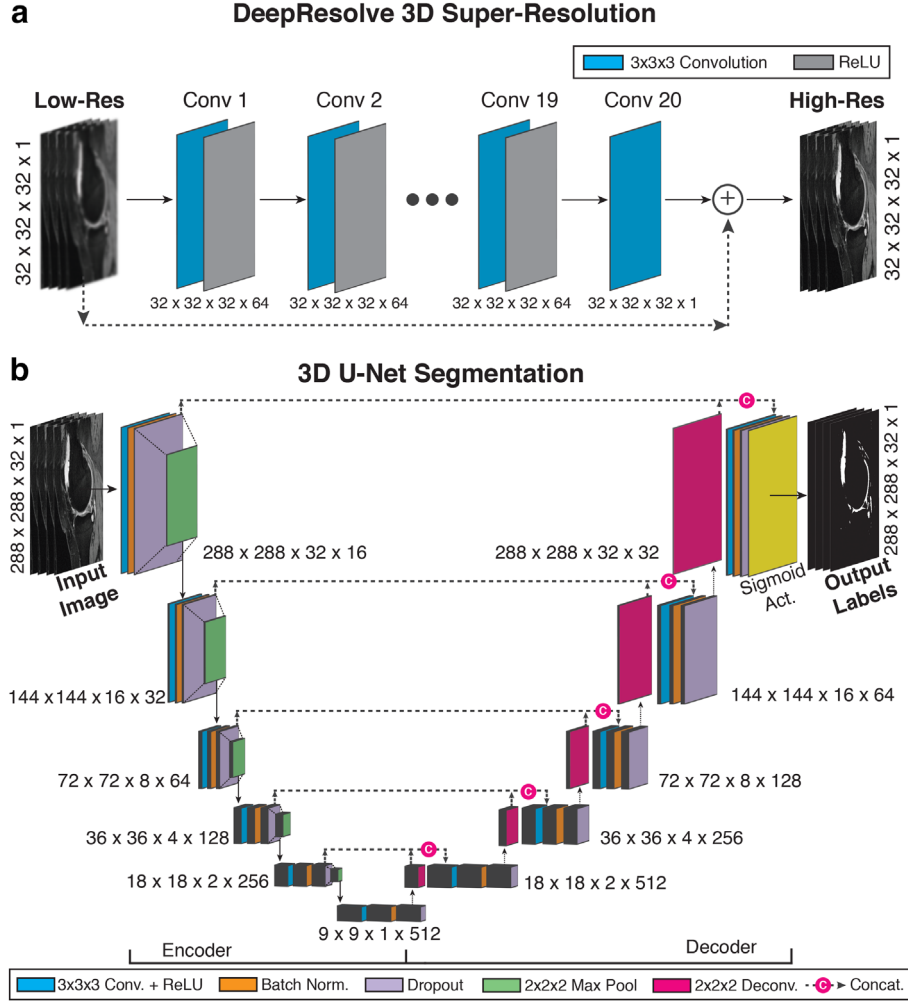


FIGURE 1: The neural network architecture for the magnetic resonance super-resolution method (DeepResolve) includes 20 3D convolutional (with 64 filters each) and rectified linear unit activation blocks (except no activation for the final convolutional layer). DeepResolve converts a low-resolution image into a residual image, where a summation of the two can create the SR image (a). The femoral cartilage segmentation network utilizes a 3D U-Net autoencoder approach consisting of six encoding and decoding convolutional layers, with filter sizes growing exponential from 32 to 512 (b).

Quantitative Cartilage Morphometry

We assessed potential blurring induced by DeepResolve by evaluating variations in cartilage morphometry, with the hypothesis that similar-appearing images should produce the same cartilage segmentation results. However, manual cartilage segmentation is a challenging task with intrareader and interreader discrepancies of $\sim 2\text{--}4\%$ coefficient of variation.⁵ Such variability may not be adequate to decouple variations in cartilage segmentation caused by image blurring or by human variability. Consequently, we designed an additional 3D CNN to perform highly accurate cartilage segmentation of the original high-resolution, DeepResolve, and TCI images in order to eliminate human interreader variability.

Inspired by previous successful approaches to perform cartilage segmentation, we designed a U-Net CNN based on the efficient encoder-decoder approach.^{26–28} However, previous approaches have utilized 2D CNNs, which may not be adequate to capture the through-plane variations in image quality of DeepResolve. As a result, we extended the current state-of-the-art 2D segmentation models to 3D. The 3D U-Net CNN utilized in this study included five encoding-decoding steps using filters increasing exponentially from

32 to 512, each with kernel size $3 \times 3 \times 3$ and rectified linear unit activations (Fig. 1b). Ground-truth segmented labels were obtained from the OAI. All slices were downsampled by a factor of 2 to increase SNR and reduce computational complexity, based on the recommendations that ~ 1.5 mm slices are adequate for cartilage morphometry.²⁹ Network training on the original high-resolution images was performed on input image dimensions of $288 \times 288 \times 32$ using a soft Dice-coefficient loss function on identical training, validation, and testing splits as the DeepResolve training.

Dice coefficients (DC), volumetric overlap error (VOE), absolute volume difference (VD), and the cartilage volume root-mean-square coefficient of variation (RMS-CV%) were calculated to assess segmentation accuracy of the network for the high-resolution, DeepResolve, and TCI images compared with the ground-truth manual labels. In addition, the same accuracy metrics were compared for DeepResolve and TCI segmentations with the original high-resolution images serving as the ground-truth. The hypothesis behind this experiment was that if DeepResolve maintained identical image quality as the original high-resolution data, it would have the exact segmentation overlap as the original images, and any

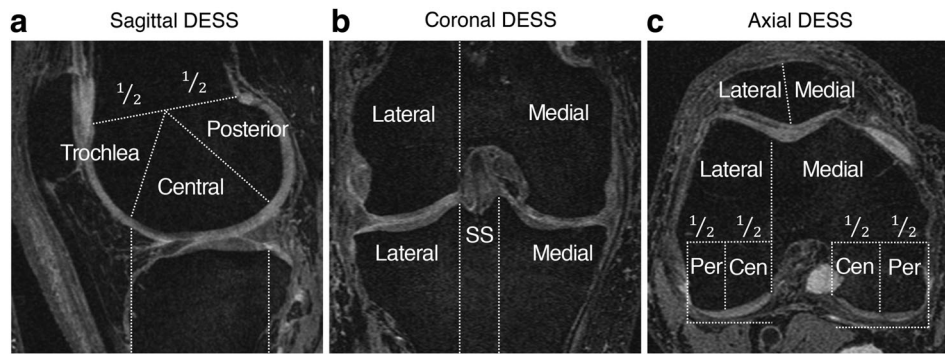


FIGURE 2: The subdivisions for the analysis of osteophytes according to the MRI Osteoarthritis Knee Score (MOAKS) criteria. In addition to the originally proposed 12 subregions, we scored two additional regions by subdividing the posterior femoral condyle into central (Cen) and peripheral (Per) compartments. The remaining subdivisions were consistent with the original MOAKS definitions. The femoral trochlea, central, and posterior edges are defined using vertical lines arising from the anterior and posterior tibial aspects. The subspinous notch (SS) and the patellar crista are defined as parts of the medial compartment. Osteophytes of the patella are scored on the superior and inferior poles.

subsequent variations in the overlap would signify the extent of image variability.

Neural Network Training

Network training for both 3D DeepResolve and 3D U-Net was performed using Keras and a Tensorflow backend (Google, Mountain View, CA) with an Adam optimizer (default parameters of $\beta_1 = 0.99$, $\beta_2 = 0.995$, $\epsilon = 1e-08$).^{30,31} A static learning rate of 0.0001 was used for DeepResolve training over 20 epochs, while a dynamic learning rate was used for U-Net (initially 0.01 decaying by 50% every four epochs for 40 epochs). DeepResolve and U-Net training was performed on NVIDIA 1080 Ti and NVIDIA Titan Xp graphical processing units (NVIDIA, Santa Clara, CA), respectively. The models that resulted in the best loss for the 35 validation datasets were chosen as the final models to be used during inference.

Reader Study

Qualitative cartilage image quality was assessed by two musculoskeletal radiologists (K.S. and J.W.) with varying levels of experience (K.S., 20 years, J.W., 3 years) and one musculoskeletal radiology resident (Am.C.). All images were presented to the radiologists in a fully blinded and randomized manner. A minimum of a 1-week washout period was maintained between reading images from the same subject to minimize memory bias. No additional information apart from the images was provided to the readers.

All three readers scored the three sets of images (original high-resolution, DeepResolve, and TCI) for sharpness, contrast, SNR, and artifacts, specifically for articular cartilage. A scoring of overall image sharpness was also performed. Scoring was performed on a 1–5 Likert scale (1 = nondiagnostic, 2 = limited quality, 3 = minimum diagnostic quality, 4 = good, 5 = high-quality). In addition to cartilage quality, the radiologists also used the DESS sequence to locate and quantify the size osteophytes in 14 subregions of the knee (Table 1). Osteophyte scoring guidelines were based on the recommendations provided in the assessment of the MRI Osteoarthritis Knee Score (MOAKS), with two modifications.³² The original MOAKS suggests scoring for osteophytes in 12 subregions of the knee; however, in this study we added two more subregions for analysis—the central and peripheral posterior femoral condyles (Fig. 2). Second, the original MOAKS guidelines suggested scoring osteophyte sizes on a scale of

0–3, without an established criteria for differentiated readings. In this study, we assigned osteophyte scores by measuring the distance the osteophyte projected beyond the joint margin where grades: 1 = 0–2 mm projection, 2 = 2–4 mm projection, and 3 = 4 + mm projection.

Referenceless Blur Factor

Most quantitative image quality metrics such as SSIM, RMSE, and pSNR rely on a high-quality reference and a lower-quality testing image. However, in many real-life scenarios a high-quality reference is not always available, which necessitates a referenceless image quality metric. Towards this end, we utilized a referenceless "blur factor" metric that was used to estimate the extent of image blurring on a single-image basis. The blur factor was proposed for evaluating natural images and has previously been used in MRI also.^{33,34} This metric applies a low-pass filter to the test image and calculates the differences between the filtered image and the original image, normalized to the intensity of the original. Sharp images would cause larger differences between the original and filtered images, compared with images that were blurry to begin with. Due to normalization, the blur factor exists in a range of 0 to 1, with 0 being least blurry and 1 being most blurry. Since blur factor relies on low-pass filtering of images where the filter kernels can be decoupled into different directions, we calculated the blurring in all three image dimensions of the original high-resolution, DeepResolve, and TCI scans. One-dimensional separable Shah functions of length 9 were used as blurring functions in all three dimensions.

Statistical Analysis

Mann–Whitney *U*-tests ($\alpha = 0.05$) with Bonferroni corrections compared the 3D U-Net segmentation performance (DC, VOE, and VD) metrics along with the blur factors for the three image sets. The same tests were also used to compare the segmentation overlap that existed between the DeepResolve and TCI segmentations using the original high-resolution segmentations as a reference. In the reader study, Mann–Whitney *U*-tests ($\alpha = 0.05$) with Bonferroni corrections also assessed variations between quality scores between the three image sets. Fleiss' kappa (κ) was used to measure overall interreader concordance, while linearly-weighted Cohen's kappa (κ) was used to measure pairwise concordance between the three readers for cartilage image quality and overall image sharpness readings.

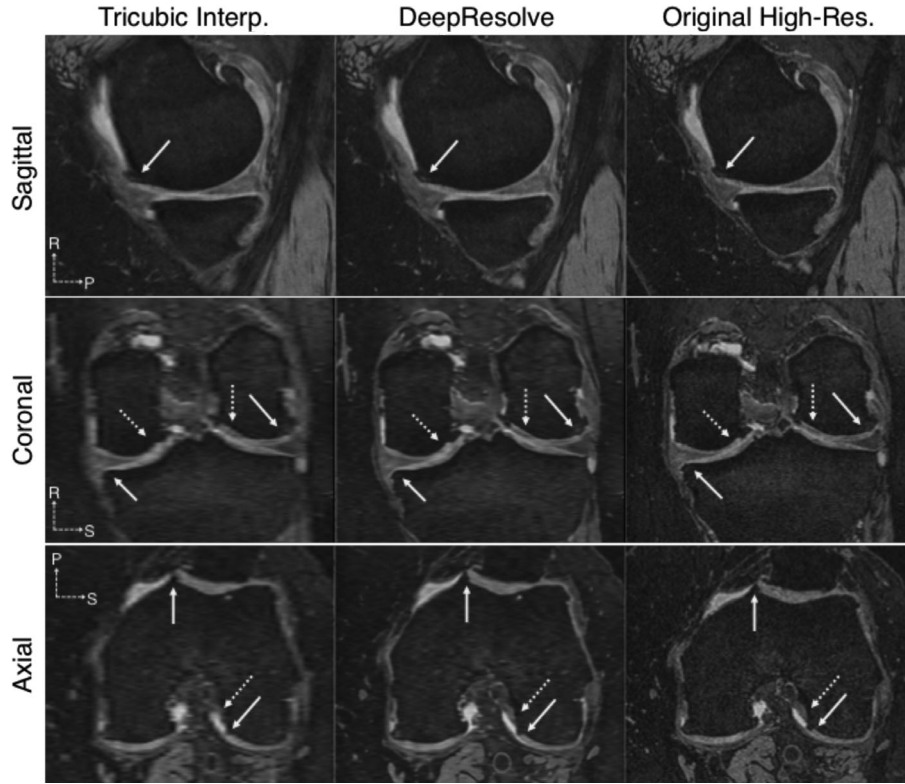


FIGURE 3: Example multiplanar reformations for the original high-resolution images, DeepResolve, and tricubic interpolated (TCI) images. Image acquisition directions are depicted as follows: readout (R), phase-encoding (P), and slice-encoding (S). Examples of osteophytes (solid arrows) in the femoral trochlea (sagittal and axial slices) and in the medial tibia (coronal slice) show that, compared with the original images, DeepResolve has high image fidelity but TCI images blur out osteophyte detail. Additionally, small cartilage features (dotted arrows) in the posterior lateral femoral condyle (coronal and axial) that are depicted well on the original and DeepResolve images are blurred out in TCI images, which affects quantitative segmentation accuracy.

For the osteophyte analysis, sensitivity, specificity, accuracy, and their corresponding 95% confidence intervals (CIs) were calculated for the DeepResolve and TCI images, using the original high-resolution as the reference standard. Using the aforementioned metrics, a diagnostic odds ratio (DOR) and its CI was calculated and tested for DeepResolve and TCI. Cochran–Mantel–Haenszel tests stratified by DeepResolve and TCI scans were also used to assess variations in osteophyte detection. Fleiss’ κ measured interreader concordance for osteophyte detection, while Cohen’s κ measured intrareader concordance comparing the DeepResolve and TCI sequences to the original high-resolution sequence. All statistical analysis was performed in Python (v. 3.6.1) using the NumPy (v1.12.1) and SciPy (v0.19.1) libraries.

Results

Example comparisons between multiplanar reformations of the original, DeepResolve, and TCI images (Fig. 3) showed that, compared with the original, DeepResolve maintains adequate image quality. However, TCI images caused a considerable blurring, which likely affected quantitative segmentation accuracy (dotted arrows, Fig. 3) and the visualization of bony osteophytes (solid arrows, Fig. 3).

Quantitative Cartilage Morphometry

The 3D U-Net network generated DC, VOE, and VD values of $90.2 \pm 1.7\%$, $17.8 \pm 2.9\%$, and $3.8 \pm 3.7\%$, respectively,

for the original high-resolution images (Fig. 4a–c). The DC, VOE, and VD quantitative DeepResolve values of $89.6 \pm 2.0\%$, $18.8 \pm 3.2\%$, and $4.6 \pm 3.9\%$ were not statistically significant compared with the original ($P = 0.38$, $P = 0.38$, and $P = 0.08$, respectively). The DC, VOE, and VD quantitative TCI values of $86.3 \pm 5.6\%$, $24.0 \pm 4.7\%$, and $5.8 \pm 5.1\%$ were statistically significant for DC ($P < 0.001$) and VOE ($P < 0.001$), but not VD ($P = 0.28$) compared with the original. The RMS-CV% values for the original high-resolution, DeepResolve, and TCI compared with the ground-truth labels was 3.1%, 2.8%, and 4.9%, respectively.

In comparisons of the segmentation overlap (Fig. 4d–f) of the DeepResolve and TCI with respect to the original scans, DeepResolve had DC, VOE, and VD values of $97.6 \pm 0.7\%$, $4.7 \pm 1.4\%$, and $1.4 \pm 1.0\%$, while TCI had values $95.0 \pm 1.1\%$, $9.5 \pm 2.0\%$, and $1.5 \pm 1.0\%$, respectively. For these comparisons, the DC and VOE metrics were significantly different ($P < 0.00001$) between the two scans, but not VD ($P = 0.33$). The DeepResolve and TCI RMS-CV % for overall cartilage volume were 0.4% and 2.6%, respectively, compared with the original.

Reader Study: Image Quality

For the reader study assessing variations in cartilage quality, DeepResolve images consistently had better performance than

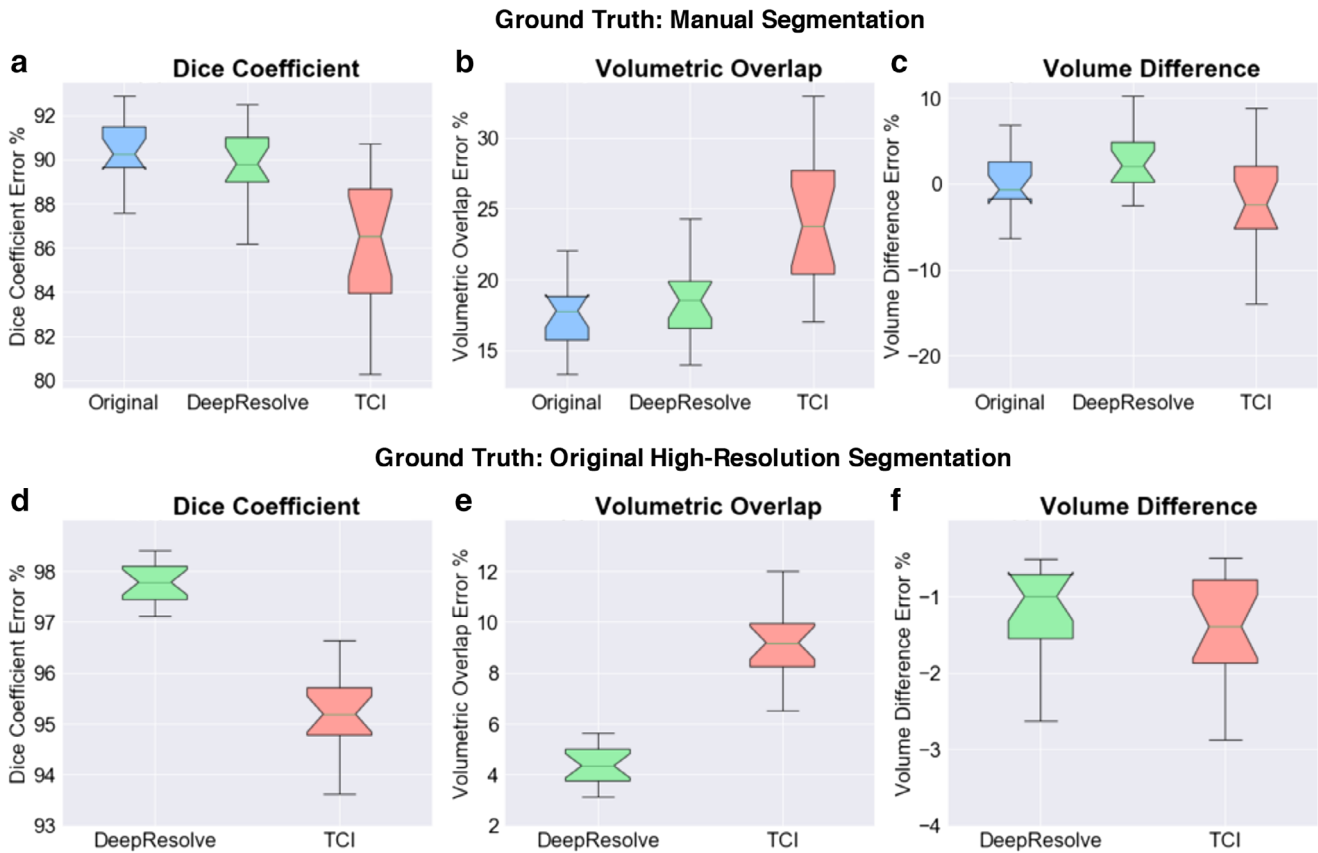


FIGURE 4: The 3D U-Net segmentation network (a–c) demonstrates accurate segmentation metrics of Dice coefficient (DC), volumetric overlap error (VOE), and volume difference (VD) values for the original high-resolution and DeepResolve images, using manual segmentations as the ground truth. TCI had significantly lower DC and VOE than DeepResolve ($P < 0.001$). Using the cartilage surface generated by segmenting the original high-resolution dataset as the reference, DC, VOE, and VD metrics were calculated for DeepResolve and TCI images. DeepResolve had significantly higher overlap ($P < 0.0001$) with the original cartilage surface than TCI.

the TCI images for all three readers (Fig. 5a–c). Following pooling of scores from the three readers (Fig. 5d), compared with TCI, both DeepResolve and the original high-resolution had significantly higher cartilage sharpness scores ($P < 1e-9$) and contrast scores ($P < 0.001$). The original high-resolution had significantly higher cartilage sharpness than DeepResolve ($P < 0.001$). All DeepResolve image quality metrics maintained a minimum diagnostic quality score of 3, unlike TCI, which had a sharpness metric of 2.3 ± 0.5 .

Interreader cartilage quality readings had a Fleiss $\kappa = 0.20$ (observed agreement = 0.54, expected agreement = 0.42). Readers 1 and 2 had a $\kappa = 0.17$ (0.05–0.28), readers 1 and 3 had a $\kappa = 0.09$ (0–0.20), and readers 2 and 3 had a $\kappa = 0.52$ (0.41–0.64). While there was only slight agreement between pairs of two readers, each reader had consistently higher scores for DeepResolve than TCI (contingency Table 2 and Fig. 5a–c). Overall image sharpness ratings for the original high-resolution images was 4.2 ± 0.5 , for the DeepResolve images was 3.7 ± 0.8 , and for the TCI images was 2.7 ± 0.9 . The original resolution images had significantly higher overall sharpness ratings than DeepResolve and TCI images ($P < 0.0001$), while the DeepResolve images also had significantly higher overall sharpness than TCI ($P < 0.0001$).

Reader Study: Osteophyte Detection

The sensitivity, specificity, and accuracy of the DeepResolve scans was higher for all osteophyte grades compared with TCI (Table 3). The overall accuracy for DeepResolve was significantly higher ($P < 0.001$) than TCI as assessed using the Cochran–Mantel–Haenszel test. The DeepResolve DOR was 32.0 (22.3–45.9) while that of TCI was 16.8 (13.0–21.8), indicating a significant ($P < 0.01$) variation between osteophyte detection for DeepResolve and TCI. DeepResolve had a κ of 0.72 (0.68–0.77), while TCI had a κ of 0.63 (0.58–0.68), compared with the original high-resolution for osteophyte detection. The overall Fleiss' κ was 0.61 (observed agreement = 0.75, expected agreement = 0.37).

Referenceless Blur Factor

The blur factors for the in-plane dimensions (x and y) were similar for all three sets of images, while there was a larger difference between the blur factors in the through-plane dimension (z) (Fig. 6). The variances of blur factors were small, suggesting repeatable behavior of the metric for images with similar preprocessing and postprocessing pipelines. While the DeepResolve z-blur factor was not the same compared with the original images, it was significantly ($P < 0.001$) superior

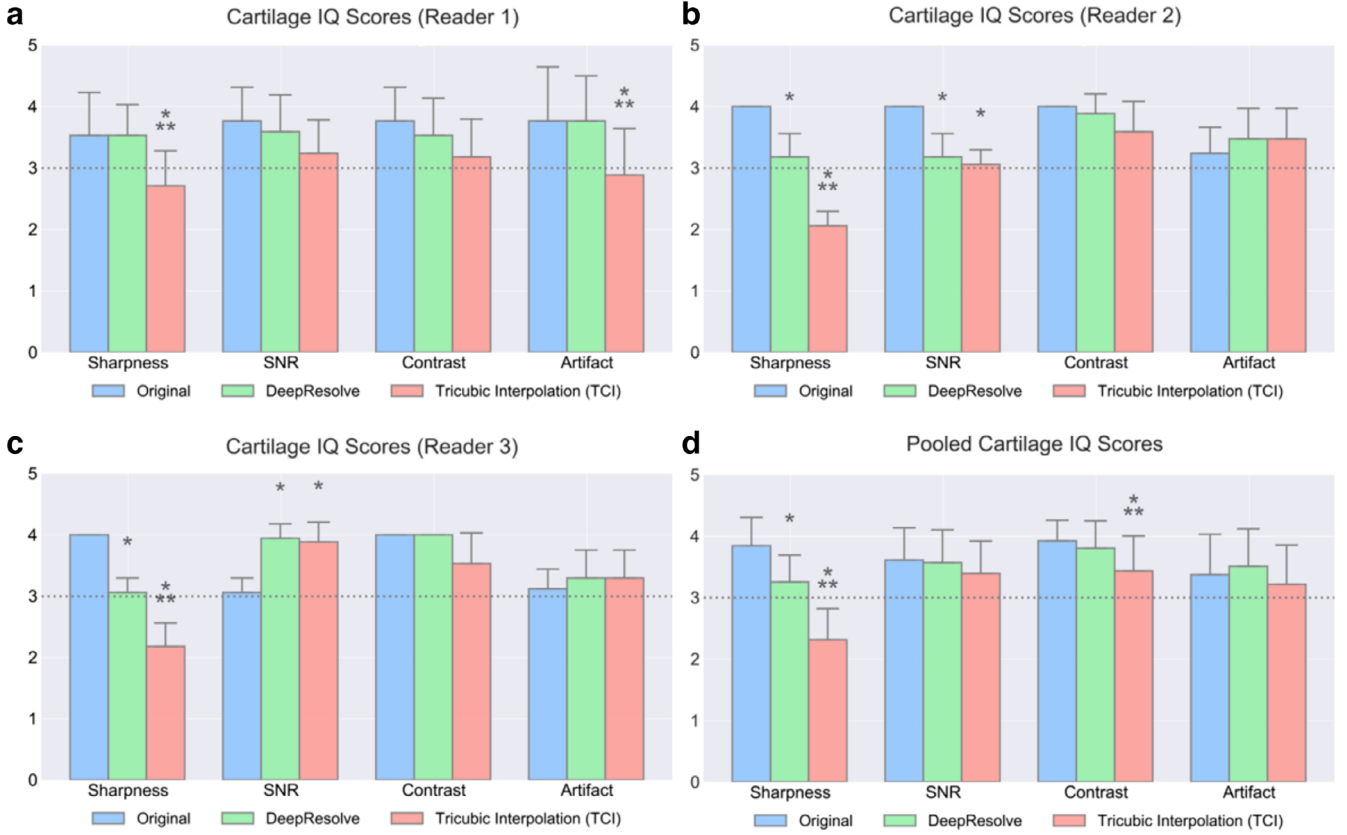


FIGURE 5: Reader 1 found TCI sharpness and artifacts significantly worse than the original and DeepResolve images (a). Readers 2 and 3 had significant sharpness differences between TCI and both the original and DeepResolve images (b,c). They also perceived significant sharpness differences between DeepResolve and original resolution images, as well as SNR differences between the original images and both, the DeepResolve and TCI images. Overall, both DeepResolve and the original images had significantly better sharpness and contrast scores, compared with TCI, while the original images had significantly better sharpness than DeepResolve (d). *Significant ($P < 0.05$) differences compared with the original high-resolution. **Significant ($P < 0.05$) differences compared with DeepResolve.

than the TCI images, corroborated using the example multi-planar reformations shown in Fig. 3. Additional examples depicting the blurring in the TCI images that can be improved using DeepResolve for enhancing the visualization of cartilage and osteophytes in the coronal and axial planes are shown in Fig. 7.

Discussion

In this study we demonstrated the utility of deep-learning-based SR beyond the assessment of image quality using quantitative SSIM, pSNR, and RMSE. We first utilized a MRI SR method (DeepResolve) to enhance the slice resolution of lower-resolution DESS scans with 3-fold thicker slices, without biasing the quantitative OA biomarkers the original high-resolution DESS sequence was utilized for. To quantify variations caused by potential blurring of the cartilage, a highly accurate fully-automated convolutional neural network demonstrated nearly identical quantitative segmentation metrics for the original high-resolution images and the DeepResolve images, both considerably better than TCI images. In a reader study evaluating qualitative image quality as assessed by

radiologists, DeepResolve significantly enhanced image quality compared with TCI. Additionally, DeepResolve also significantly outperformed TCI in the detection of osteophytes. We also demonstrated the utility of the blur factor metric to ascertain a referenceless measure of image blurring. DeepResolve did not match the high image quality of the original high-resolution data, but was able to considerably outperform TCI in image quality and biomarker accuracy. This finding may be promising, since several MRI visualization and analysis techniques rely on the use of interpolation methods to resize images. Moreover, such results demonstrate that SR is not only a promising method for enhancing image quality, but it can also be used to recreate quantitative biomarkers that necessitate high-resolution MRI.

The 3D U-Net segmentation network was able to generate very high accuracy for performing automated femoral segmentation. Using such an automated method that has an accuracy coefficient of variation similar to the variability between two human readers eliminated human uncertainty during cartilage segmentation.^{5,29} The use of such a network to demonstrate a high overlap between the original high-resolution segmented volume and the DeepResolve

TABLE 1. Osteophyte Scoring Paradigm Which Summarizes the Locations That Osteophytes Along With a Criteria for Determining Osteophyte Grade, Based on the Extent of Protrusion From the Joint Surface

Bone	Region	Side	Scan plane
Femur	Trochlea	Medial	Axial
		Lateral	Axial
	Posterior	Medial peripheral	Axial
		Medial central	Axial
		Lateral peripheral	Axial
		Lateral central	Axial
	Central	Medial	Coronal
Patella	N/A	Lateral	Coronal
		Superior	Sagittal
	N/A	Inferior	Sagittal
		Medial	Axial
		Lateral	Axial
Tibia	N/A	Medial	Coronal
		Lateral	Coronal
Protrusion	0–2 mm	2–4 mm	4+ mm
Osteophyte Grade	1	2	3

segmented volume depicted that the neural network perceived both sets of images relatively similarly. Exactly similar images fed through the 3D U-Net would produce overlap metrics of 100% DC, and 0% VOE, VD, and CV% values, which were similar to the actual results generated. In comparison, traditional methods such as TCI performed significantly worse, which was expected since interpolation has been shown to blur out subtle MRI features.³⁵

While the 3D U-Net evaluated image quality from a quantitative metric, the reader study was able to determine subjective image quality specific to tissues of interest. The DeepResolve metrics of contrast and artifacts were comparable to that of the original high resolution. The DeepResolve sharpness metric was slightly lower than the high resolution, but it was considerably higher than TCI, which is a commonly used technique. There was, however, a low concordance among the three readers, but this was primarily caused by variations in grading the quality metrics either 3 (minimum diagnostic quality) or 4 (good quality), which can be challenging to distinguish between, especially when only analyzing subtle tissues such as cartilage. While Likert scales are

TABLE 2. Contingency Tables Depicting the Interreader Variations Between Pairs of the Three Readers in the Analysis of Cartilage Image Quality

		Reader 2				
Cartilage Image Quality		2	3	4	5	Total
Reader 1	2	6	6	7	0	19
	3	9	39	39	0	87
	4	1	34	53	0	88
	5	0	6	4	0	10
	Total	16	85	103	0	204

		Reader 3				
Cartilage Image Quality		2	3	4	5	Total
Reader 1	2	5	7	7	0	19
	3	8	34	45	0	87
	4	1	38	49	0	88
	5	0	6	4	0	10
	Total	14	85	105	0	204

		Reader 3				
Cartilage Image Quality		2	3	4	5	Total
Reader 2	2	14	0	0	0	14
	3	2	54	29	0	85
	4	0	31	74	0	105
	5	0	0	0	0	0
	Total	16	85	103	0	204

All readings are pooled across the readings of sharpness, signal-to-noise ratio, contrast, and artifacts for 17 subjects assessed using all three image sets (original high-resolution, DeepResolve super-resolution, and tricubic interpolation). Note that no scores of 1 (nondiagnostic) were assigned.

based on subjective readings that can be affected by reader experience and comfort levels, there was a general and consistent trend of DeepResolve consistently outperforming TCI stayed for all three readers.

The reader study also demonstrated that the DeepResolve sequence had higher conspicuity for detecting osteophytes as compared with TCI. Identifying osteophytes requires the ability to perform multiplanar reformations, which requires a high in-plane and through-plane resolution, making it an ideal application area for through-plane slice resolution. The sensitivity for detecting osteophytes was consistent over all osteophyte grades, while the DeepResolve sensitivities for subtle osteophytes (grades 1 and 2) were considerably higher than those of TCI. The specificity was the

TABLE 3. Sensitivity, Specificity, and Accuracy and the Corresponding Confidence Intervals (in Parentheses) for Grading Osteophytes on a Scale of 0–3 Using the DeepResolve and TCI Images, With the Original High-Resolution Images as the Reference Standard

Osteophyte Grade	Incidence	Sensitivity			Specificity			Accuracy	
		DeepResolve	TCI		DeepResolve	TCI		DeepResolve	TCI
1	270	0.80 (0.74–0.84)	0.73 (0.68–0.89)		0.88 (0.85–0.91)	0.85 (0.82–0.89)		0.85 (0.82–0.88)	0.81 (0.78–0.84)
2	332	0.79 (0.74–0.83)	0.70 (0.65–0.75)		0.83 (0.79–0.86)	0.78 (0.73–0.82)		0.81 (0.78–0.84)	0.74 (0.71–0.78)
3	109	0.81 (0.72–0.88)	0.78 (0.69–0.85)		0.97 (0.95–0.98)	0.94 (0.91–0.95)		0.94 (0.92–0.96)	0.91 (0.89–0.93)
Total	711	0.80 (0.76–0.82)	0.73 (0.69–0.76)		0.93 (0.92–0.94)	0.91 (0.90–0.92)		0.90 (0.89–0.91)	0.86 (0.85–0.88)

All results are pooled across all patients and readers.

highest for the largest (grade 3) osteophytes for both DeepResolve and TCI scans, suggesting that if the osteophyte is large enough, it has similar conspicuity on both scans. The interreader and intrareader agreement was comparable to previous studies where the initial MOAKS scoring criteria was presented and validated by expert musculoskeletal radiologists.³² The overall high diagnostic odds ratio, inter- and intrareader agreement, and accuracy of DeepResolve suggests that SR may not bias the detection of osteophytes.

The DESS sequence was included in the OAI primarily to evaluate the morphometry of cartilage and the high resolution was later utilized to evaluate variations in bone shape.^{6,29,36} In this study, we evaluated the impact of using SR to extract the biomarkers that the original DESS enabled. The increased accuracy of cartilage and osteophyte findings, and the image quality of DeepResolve images compared with TCI images, showed that DeepResolve considerably enhanced the image quality of the TCI inputs. However, the differences between the DeepResolve and the original resolution images demonstrated that the SR method could be further improved in order to remove the residual blurring between the two sets of images. Nonetheless, using such a case study that necessitates the need for high-resolution DeepResolve may be a promising method to accelerate future MRI scans that are acquired with a lower resolution. In addition, the underlying SR neural network has been extended to other pulse sequences such as qDESS, which separates the two echoes generated in DESS and also calculates an automatic T₂ relaxation time map.^{19,37}

Accurately evaluating image quality is one of the primary challenges in deep-learning medical image reconstruction or enhancement techniques. Reductions in traditional cost functions such as mean absolute error or mean square errors do not correspond to perceptual image quality.¹⁶ Here we proposed a novel method for quantifying the results of the DeepResolve technique, which was successful in determining the extent of blurring in the three sets of images, and conversely, the resolution enhancement of the SR methods. As depicted in the blur factor plots, the in-plane blurring (x and y directions) was primarily induced during the TCI process and, overall, DeepResolve only culminated in minimal changes in in-plane blurring compared with the original resolution images. The additional benefit of the blur factor was that it did not require a reference high-quality image for comparison, which could be beneficial for use as an optimization criteria in unsupervised learning algorithms or when a reference image is unavailable. The low-pass filter used in this study utilized a 1D Shah function (of length 9) as per previous recommendations; however, future studies could investigate the use of additional blurring filters. Additionally, in this work slice resolution enhancement was chosen because most MRI vendors have slice-interpolation options available and because musculoskeletal MRI is normally performed with 2D fast spin echo sequences that have a high in-plane resolution,

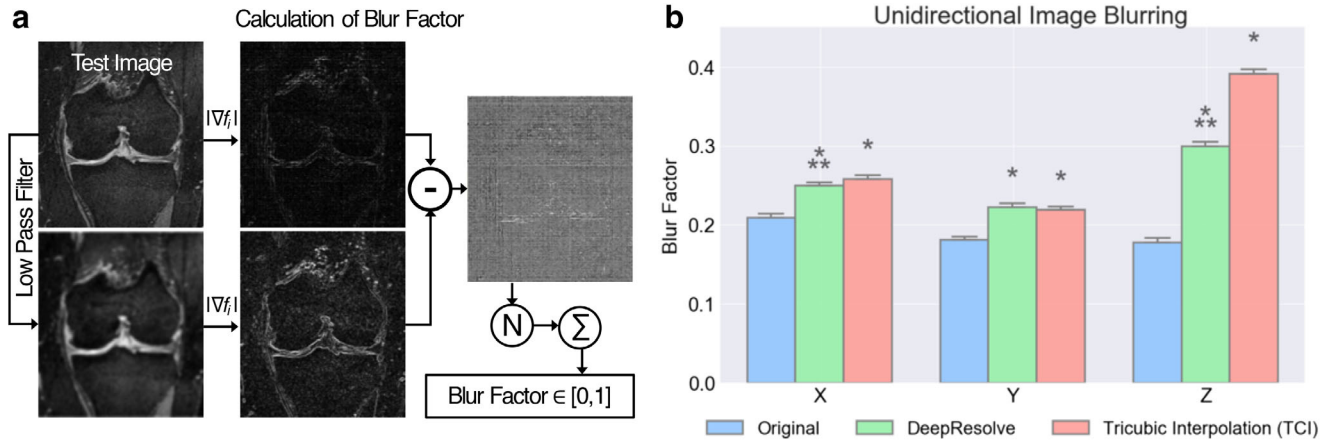


FIGURE 6: The referenceless blur factor is calculated by convolving the test image with a blurring kernel and evaluating the absolute value of the image gradients ($|\nabla f|$) that are normalized (denoted by "N") to the original image intensity (a). The blur factors demonstrate minimal blurring variations between the three scans in the "x" and "y" directions (in-plane), but the blurring in the "z" (through-plane) was considerably different between scans (b). TCI had the worst through-plane blurring, while DeepResolve was able to start with TCI images and enhance their quality to reduce the through-plane blurring. *Significant ($P < 0.05$) differences compared with the original high-resolution. **Significant ($P < 0.05$) differences compared with DeepResolve.

but thicker slices and slice gaps. However, future work will be necessary to investigate the tradeoffs with downsampling in different dimensions in order to determine the ideal technique for limiting scan time.

While the results from this study demonstrate the promise of DeepResolve, this study had several limitations. First, the analysis of the quantitative cartilage morphometry was performed using an additional CNN rather than a trained human

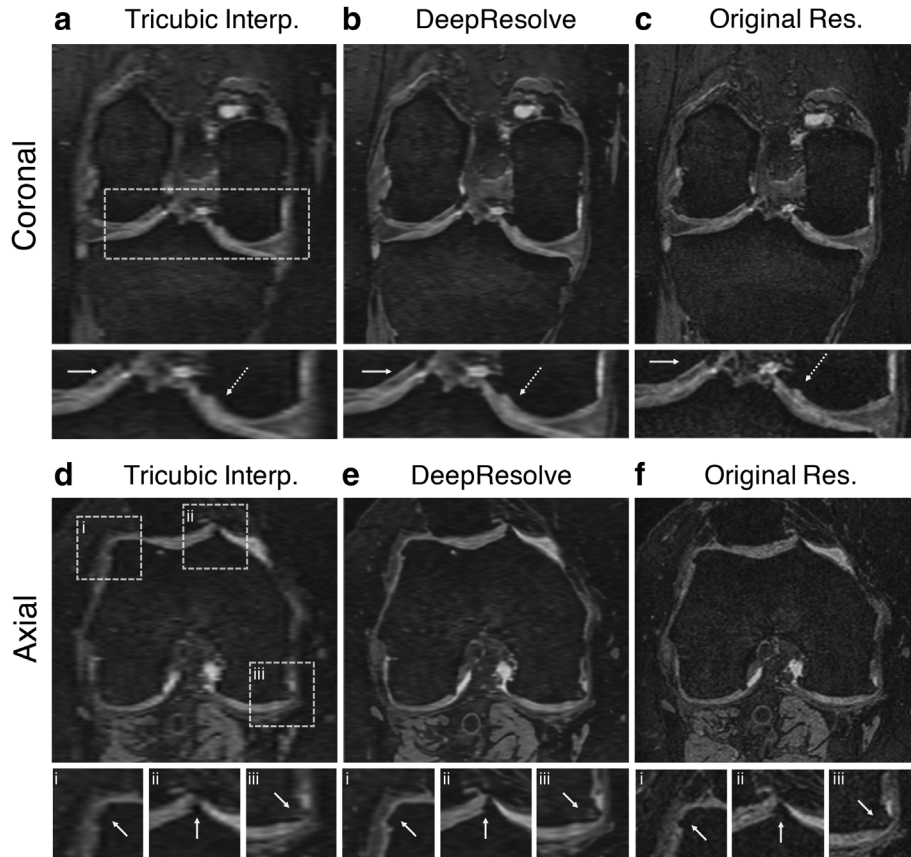


FIGURE 7: Example coronal reformats (a–c) demonstrate the variations in the depiction of cartilage using the three image sets. Using the zoomed in inlays (dotted box), DeepResolve enhanced the appearance of jagged cartilage artifacts (solid arrow) and sharpened the contours of the cartilage–bone interface (dotted arrow), compared with TCI. Similarly, example axial reformats (d–f) and the zoomed in inlays show that DeepResolve enhanced the depiction of several osteophytes whose contours were blurred out in TCI images. The depiction of both cartilage and osteophytes using DeepResolve had a closer resemblance to the sharper original resolution images than the blurrier TCI images.

observer. However, since the expected variation between the original high-resolution and the DeepResolve scans was subtle, it would have been challenging to decouple human segmentation variations from the interscan variations. While the reader study assessed cartilage image quality, an analysis of cartilage lesions may have been useful. However, the DESS sequence used in the OAI has low sensitivity to cartilage lesions, likely due to the combination of the two separate DESS echo contrasts.^{38,39} Future studies separating the two DESS echoes, akin to qDESS, may provide an improved estimate of structural abnormalities. Third, the analysis of osteophytes was binned into discrete 0–3 categories instead of reporting the actual size due to the challenges in standardizing readings between different readers. Future studies could potentially perform segmentation of the osteophytes to evaluate whether SR changes their perceived shape and dimensions. Moreover, applying these findings to larger cohorts would be beneficial to determine the utility of SR, especially for subjects suffering from varying OA severities. Our preliminary findings also suggest that DeepResolve (trained on Siemens single-contrast DESS images) can be fine-tuned to enhance GE multicontrast quantitative DESS images (that generates two varied contrasts) using only 30 patients.¹⁹ However, additional characterization ascertaining the generalizability of DeepResolve to additional sequences, protocols, and vendors will be necessary for widespread adoption.

In conclusion, in an effort to interpret the results of a deep-learning-based SR neural network, we have quantitatively shown that SR minimally affects perceived global image blurring, and qualitatively and quantitatively shown that it minimally biases cartilage and osteophyte biomarkers and image quality. Based on such a performance that minimally blurs or biases subtle musculoskeletal tissues, SR may be a more promising technique than naïve interpolation for accelerating image acquisition by transforming low-resolution images that can be acquired rapidly into higher-resolution images.

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Disclosures

A.C. has provided consulting services to Skope MR Inc, Subtle Medical, and Chondrometrics GmbH; and is a shareholder of Subtle Medical, LVIS Corporation, and Brain Key. Z.F. is an employee of LVIS Corporation. G.G. and B.H. have received research support from GE Healthcare and Philips. B.H. is a shareholder of LVIS Corporation. Neither organizations were involved in the design, execution, data analysis, or the reporting of this study.

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